

Finite Source Signatures, Joint Incidence, and Representation Uniqueness

Tau Core Foundation Paper II

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Draft v0.2 — 2026

Abstract

We study a finite source-selection problem that precedes terminal physical readouts. A complete typed source signature consists of a finite-dimensional unital C^* -algebra, a faithful source state, multiplication and involution data, and distinguished role elements or completely positive maps. The GNS construction then gives a unique minimal cyclic realization up to the typed unitary equivalences already declared in the source packet.

This positive theorem does not make terminal marginals sufficient. Two faithful bipartite states can share both marginals and differ in a joint moment. More strongly, four-qubit states can share every proper marginal while differing along the 81-dimensional weight-four Pauli subspace; proper marginals span only 174 of 255 traceless directions. A single source-owned onto joint incidence with positive stiffness is sufficient to generate a faithful joint Gram state and hence all finite mixed moments. Equivalently, one coherent source homomorphism Π_{phys} selects the physical relation ideal and its faithful quotient representation.

We extend the four-qubit count to arbitrary n -partite qudit systems and give singular-value bounds that quantify how an onto incidence becomes ill-conditioned before rank is lost. The results characterize sufficient finite source information and exact non-identifiability boundaries. They do not prove that a physical base-seed parent creates, selects, or occupies the required joint incidence or Π_{phys} , and they do not derive terminal physics, the Standard Model, or empirical validation. This is the second part of a three-paper foundation sequence: Paper I freezes the architecture and conditional completion, while Paper III analyzes record transport and recovery after a source packet has been fixed.

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graph LR
    P1[Paper I  
architecture + conditional completion] --> P2[Paper II  
finite source sufficiency + incidence]
    P2 --> P3[Paper III  
record transport + recovery]
    P1 -.->|shared open arrow: physical base--seed selection and occupation| P2

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Later papers refine sufficient internal conditions; they do not prove nature-level selection.

1 Problem and claim boundary

Which source data determine one minimal joint realization, and which apparently complete data remain non-identifying?

Series notation ledger.

Symbol	Series-wide meaning	Primary treatment
s	source/seed packet prior to observer descent	Paper I
M_τ	stabilized morphological body	Paper I
Desc_O	observer-indexed physical descent/readout	Paper I
Π_{phys}	selected coherent source representation	Paper II
x_A	normalized finite source/body action separation	Papers I, III
x_Q	observer-carrier Uhlmann chord separation	Papers I, III
$\mathcal{C}, \mathcal{R}$	record map and recovery map	Paper III

Let $\mathcal{A}_s$ be a finite-dimensional unital C^* -algebra. After the declared source-null quotient, assume a positive stiffness $K_* > 0$ and a deduplicated source incidence R_s^{dedup} . On its occupied ideal, define

$$G_s = R_s^{\text{dedup}} K_*^{-1} (R_s^{\text{dedup}})^\dagger, \quad \rho_s = \frac{G_s}{\text{Tr } G_s}, \quad \omega_s(a) = \text{Tr}(\rho_s a). \quad (1)$$

Choose a source-orthonormal basis $e_1, \dots, e_d$. Let $\mathbf{z}_s$ be the finite list of distinguished typed role elements and CP maps retained by the claim.

Finite source-signature logic

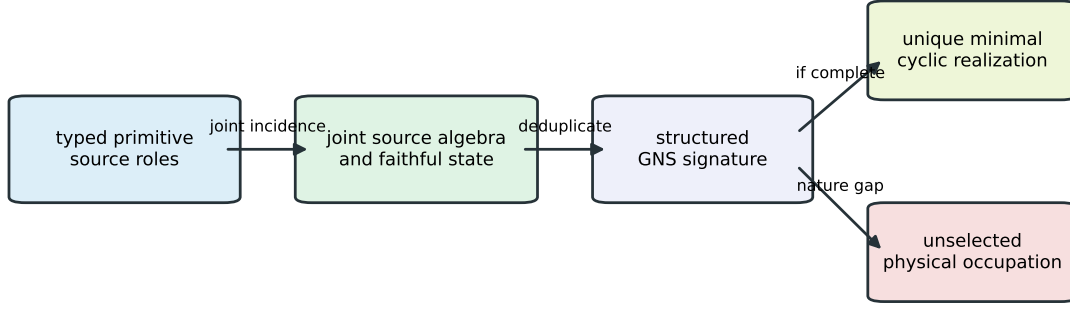


Figure 2: The finite logic. Complete source data determine a minimal cyclic realization, but the physical base-seed law has not yet been shown to create or occupy those data.

Definition 1 (Complete finite typed signature). *The source signature is*

$$\Sigma_s = \left(c_{ab}^c, j_a^b, \omega_s(e_a), [z_\alpha]_a, [\mathcal{I}_\beta]_{ab} \right), \quad e_a e_b = c_{ab}^c e_c, \quad e_a^* = j_a^b e_b. \quad (2)$$

It is complete relative to a claim when it contains the full abstract algebra, faithful state and every typed element or map used by that claim. Simultaneous source-basis changes are quotiented only when declared typed source isometries. Undeclared algebra automorphisms are not automatically physical equivalences.

This is finite information: multiplication, involution, state and role arrays. Completeness is claim-relative, not a statement that all parent information is accessible.

Every finite-dimensional unital C^* -algebra has a decomposition

$$\mathcal{A}_s \cong \bigoplus_{\alpha=1}^r M_{n_\alpha}(\mathbb{C}). \quad (3)$$

The signature must therefore record not only dimensions but the multiplication and involution that distinguish the blocks and their typed roles. A faithful state has strictly positive density on every occupied central block. Dropping a zero-weight block is an occupation decision; it is not a consequence of GNS uniqueness on the remaining support.

2.1 Four logically distinct questions

The following distinctions are used throughout.

Question	Mathematical content	What can answer it here?
Existence	Does at least one realization of the typed packet exist?	An explicit representation or incidence construction.
Uniqueness	Are all minimal cyclic realizations of the same complete packet typed-unitarily equivalent?	The finite GNS theorem below.
Identifiability	Do the retained observables determine the complete packet?	A separating-rank test, or a countermodel in the kernel of the retained descriptor.

Question	Mathematical content	What can answer it here?
Physical tion	selec- Why is one admissible packet oc- cupied by Nature?	A base-seed source law or indepen- dent experiment; neither follows from the other three rows.

In particular, uniqueness conditional on Σ_s cannot repair an incomplete descriptor. Conversely, an informationally complete descriptor does not prove physical occupation of its reconstructed point.

Definition 2 (Typed equivalence). *Two realizations are equivalent only if a unitary intertwines the represented algebra, maps the cyclic source vector, preserves the declared central blocks, and intertwines every distinguished role element and source map. Automorphisms not included in this equivalence group remain physically distinguishable data.*

3 GNS representation uniqueness

Proposition 1 (Finite source-signature representation uniqueness). *Fix $(\mathcal{A}_s, \omega_s, \mathbf{z}_s)$ through one complete signature Σ_s , with ω_s faithful. Any two minimal cyclic finite realizations of these data are related by a unique unitary that maps one cyclic vector to the other, implements the fixed algebra identification, and intertwines every distinguished element and the represented action of every distinguished linear or completely positive source map.*

Proof. Faithfulness gives $\mathcal{N}_{\omega_s} = \{a : \omega_s(a^*a) = 0\} = \{0\}$. The GNS Hilbert space is the completion of $\mathcal{A}_s$ under $\langle a, b \rangle_s = \omega_s(a^*b)$, with cyclic vector $[1]$ and left representation $\pi_s(a)[b] = [ab]$. For two minimal cyclic realizations set

$$U\pi_1(a)\Omega_1 = \pi_2(a)\Omega_2. \quad (4)$$

Equality of ω_s makes U isometric; cyclic minimality makes its range dense, hence unitary. For a distinguished source map $\mathcal{I}_\beta : \mathcal{A}_s \rightarrow \mathcal{A}_s$, define on the cyclic domain

$$\widehat{\mathcal{I}}_{\beta,i} \pi_i(a)\Omega_i = \pi_i(\mathcal{I}_\beta(a))\Omega_i. \quad (5)$$

In finite dimension this is a bounded linear operator whenever the displayed rule is well defined on the GNS quotient; faithfulness makes that quotient trivial here. The definition of U gives $U\widehat{\mathcal{I}}_{\beta,1} = \widehat{\mathcal{I}}_{\beta,2}U$ on the cyclic span. The same calculation handles represented role elements. The cyclic span fixes U uniquely. This is the finite GNS uniqueness mechanism [1]. $\square$

Complete positivity is not used to prove unitary uniqueness; it records the physical type of an admissible source operation. Stinespring dilation gives a standard representation of such maps, but the dilation environment is not unique without its own minimality condition [2]. The present theorem therefore claims uniqueness of the typed minimal cyclic realization, not uniqueness of every implementation of its CP operations.

The theorem begins *after* the source signature has been supplied. It does not show that a physical parent selects Σ_s . If a signature map Φ_s is faithful modulo typed source isometry and the post-quotient metric is positive, then

$$\mathcal{D}_s(C) = \|\Phi_s(C) - \Sigma_s\|^2 \quad (6)$$

has at most one zero completion class. Existence of an occupied zero remains an independent physical requirement.

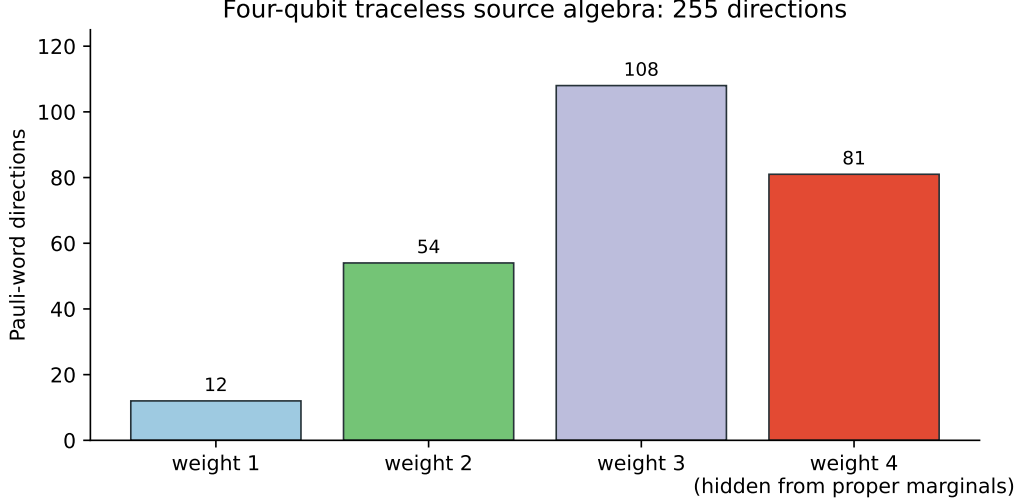


Figure 3: Exact Pauli-word dimension audit. Every proper marginal can access weights at most three; the 81 weight-four directions remain invisible.

4 Marginals do not select a joint source

Proposition 2 (Two-sector marginal no-go). *Complete faithful source states on separate sectors do not generally determine their joint source state.*

Proof. For two qubits, compare

$$\rho_0 = \frac{I_4}{4}, \quad \rho_\epsilon = \frac{1}{4} (I_4 + \epsilon \sigma_z \otimes \sigma_z), \quad 0 < |\epsilon| < 1. \quad (7)$$

Both states are faithful and both marginals equal $I_2/2$. Their joint $\sigma_z \otimes \sigma_z$ moments are respectively zero and ϵ . $\square$

Thus compatible gravity, clock, radiation, quantum, matter or gauge marginals cannot be promoted to one joint physical source merely by being placed side by side. Cross-sector moments spanning the missing self-adjoint source directions are required.

5 All proper marginals remain incomplete

For four qubits the traceless Pauli words have total dimension $4^4 - 1 = 255$. Words of weights one, two and three contribute

$$\binom{4}{1}3 + \binom{4}{2}3^2 + \binom{4}{3}3^3 = 12 + 54 + 108 = 174. \quad (8)$$

The remaining weight-four subspace has dimension $3^4 = 81$.

Proposition 3 (Four-sector all-proper-marginal no-go). *There are faithful four-qubit states with identical marginals on every proper subsystem and different genuine four-sector moments.*

Proof. Let

$$\rho_\epsilon = \frac{1}{16} (I_{16} + \epsilon \sigma_z^{\otimes 4}), \quad 0 < |\epsilon| < 1. \quad (9)$$

Its eigenvalues are $(1 \pm \epsilon)/16 > 0$. Tracing out any nonempty factor annihilates the Pauli product because $\text{Tr} \sigma_z = 0$, so every proper marginal is maximally mixed and independent of ϵ . Yet $\text{Tr}(\rho_\epsilon \sigma_z^{\otimes 4}) = \epsilon$. $\square$

The number 81 belongs specifically to this four-qubit Pauli decomposition. It is not a representation-independent count for every four-sector model. The invariant lesson is that all proper marginals may leave genuine joint directions unresolved.

5.1 General n -partite qudit count

The preceding example is the smallest calculation used by the Tau packet, but the obstruction is general. Fix a Hilbert–Schmidt orthogonal Hermitian basis $\{I, T_1, \dots, T_{d^2-1}\}$ on one qudit, with every T_a traceless. The tensor words of weight w have dimension

$$N_w = \binom{n}{w} (d^2 - 1)^w. \quad (10)$$

Proposition 4 (All-proper-marginal kernel for n qudits). *For $n \geq 2$, the linear descriptor consisting of every proper marginal annihilates the full-weight tensor subspace of dimension $(d^2 - 1)^n$. Hence the descriptor rank on traceless Hermitian operators is at most*

$$\sum_{w=1}^{n-1} \binom{n}{w} (d^2 - 1)^w = d^{2n} - 1 - (d^2 - 1)^n. \quad (11)$$

There are faithful density operators with identical proper marginals and different full-weight moments.

Proof. Tracing out any factor of a full-weight word $T_{a_1} \otimes \dots \otimes T_{a_n}$ gives zero because each local factor is traceless. Linear independence of the tensor-product basis gives the kernel dimension. For any nonzero Hermitian full-weight word X , set

$$\rho_\epsilon = \frac{I}{d^n} + \epsilon X, \quad |\epsilon| < \frac{1}{d^n \|X\|_{\text{op}}}. \quad (12)$$

Then $\rho_\epsilon > 0$, has unit trace, and every proper marginal is independent of ϵ , while $\text{Tr}(\rho_\epsilon X)$ varies linearly. $\square$

For $n = 4, d = 2$, Eq. (11) returns $255 - 81 = 174$. The theorem is an identifiability statement, not a claim that physical source sectors are qudits. It formalizes why assembling all lower order readouts need not determine their common parent relation, a familiar form of the quantum marginal problem [3, 4, 5].

5.2 Descriptor rank criterion

Let V_J be the real vector space of traceless self-adjoint operators on a declared joint carrier, and let $D : V_J \rightarrow \mathbb{R}^m$ collect all retained expectation values. The joint state is locally identifiable from those linear data exactly when

$$\ker D = \{0\}. \quad (13)$$

If $\ker D \neq \{0\}$, any sufficiently small Hermitian perturbation along a kernel direction preserves positivity around a faithful state and produces a locally indistinguishable state. This turns “descriptor completeness” into a finite rank test rather than a verbal completeness claim.

Proposition 5 (Noise-stable linear identifiability). *Equip V_J and $\mathbb{R}^m$ with fixed Euclidean norms and let $\sigma_D > 0$ be the smallest singular value of an injective descriptor D . If two states satisfy*

$$\|D(\rho - \sigma)\| \leq \delta, \quad (14)$$

then

$$\|\rho - \sigma\| \leq \frac{\delta}{\sigma_D}. \quad (15)$$

If D is not injective, no finite bound of this form can control components in $\ker D$.

Proof. By the singular-value variational definition, $\|DX\| \geq \sigma_D\|X\|$ for all $X \in V_J$. Apply this to $X = \rho - \sigma$. If $X \in \ker D \setminus \{0\}$, the descriptor difference vanishes while $\|X\|$ does not. $\square$

The bound depends on the declared normalization of observables and noise. It is not invariant under arbitrary rescaling of descriptor rows, so a physical audit must freeze those units before reporting σ_D . Statistical confidence regions may then be propagated through the same singular-value decomposition; they do not change the exact kernel.

For orientation, three qubits have $4^3 - 1 = 63$ traceless Pauli directions. All one- and two-body words span

$$\binom{3}{1}3 + \binom{3}{2}3^2 = 9 + 27 = 36, \quad (16)$$

leaving 27 weight-three directions invisible to every proper marginal. Four qubits enlarge these numbers to 174 visible and 81 invisible directions. The growth is structural: adding more lower-order precision does not reveal an exact full-weight kernel direction.

6 One onto joint incidence is sufficient

The positive replacement is not an independently fitted coefficient for every missing word.

Proposition 6 (Onto-incidence faithful-state theorem). *Let $\mathcal{H}_J$ be an occupied finite joint carrier, let $\mathcal{E}_J$ be the source-leg Hilbert space, let $K_s \in \text{End}(\mathcal{E}_J)$ be positive definite, and let $R_J : \mathcal{E}_J \rightarrow \mathcal{H}_J$ be a source-owned onto incidence. Then*

$$G_J = R_J K_s^{-1} R_J^\dagger, \quad \rho_J = \frac{G_J}{\text{Tr } G_J} \quad (17)$$

defines a faithful density operator on $\mathcal{H}_J$. Once the typed joint role algebra is represented on $\mathcal{H}_J$, it fixes every finite mixed moment of that representation.

Proof. Surjectivity of R_J makes $R_J^\dagger$ injective. For nonzero $v \in \mathcal{H}_J$,

$$\langle v, G_J v \rangle = \langle R_J^\dagger v, K_s^{-1} R_J^\dagger v \rangle > 0. \quad (18)$$

Hence $G_J > 0$, the normalized state is faithful, and linearity assigns all mixed-word expectations. $\square$

Onto incidence is sufficient, not necessary in every possible presentation. It is also not selected by the separate sector marginals.

6.1 Quantitative stability before rank loss

Surjectivity is a sharp condition, but experiments and numerical source reconstructions require a condition number. Write $k_{\min} = \lambda_{\min}(K_s)$, $k_{\max} = \lambda_{\max}(K_s)$, and let $\sigma_{\min}(R_J)$, $\sigma_{\max}(R_J)$ be the extremal singular values on the non-null source quotient.

Proposition 7 (Joint-Gram eigenvalue bounds). *If R_J is onto, then*

$$\frac{\sigma_{\min}(R_J)^2}{k_{\max}} \leq \lambda_{\min}(G_J) \leq \lambda_{\max}(G_J) \leq \frac{\sigma_{\max}(R_J)^2}{k_{\min}}. \quad (19)$$

Consequently,

$$\kappa(G_J) \leq \kappa(K_s) \left(\frac{\sigma_{\max}(R_J)}{\sigma_{\min}(R_J)} \right)^2. \quad (20)$$

Proof. For unit $v \in \mathcal{H}_J$,

$$\frac{\|R_J^\dagger v\|^2}{k_{\max}} \leq \langle v, G_J v \rangle \leq \frac{\|R_J^\dagger v\|^2}{k_{\min}}. \quad (21)$$

The min–max principle and the singular values of $R_J^\dagger$ give Eq. (19); division gives Eq. (20). $\square$

Thus the exact theorem has an observable precursor: as $\sigma_{\min}(R_J) \rightarrow 0$, the reconstructed joint state becomes poorly conditioned before faithfulness is lost. A small singular value is not new physics by itself; it may also signal an incomplete descriptor, calibration degeneracy, or insufficient data. If R_J is not onto, then

$$\ker G_J = \ker R_J^\dagger = (\text{ran } R_J)^\perp. \quad (22)$$

The unresolved directions are therefore explicit rather than absorbed into a fitted terminal correction.

7 The coherent physical source map

Let the once-owned primitive roles generate a universal typed unital $*$ -algebra $\mathcal{A}_{\text{prim}}$ modulo relations already fixed source-side. One source-owned unital $*$ -homomorphism

$$\Pi_{\text{phys}} : \mathcal{A}_{\text{prim}} \longrightarrow \mathcal{B}(\mathcal{H}_{\text{post}}) \quad (23)$$

selects the physical relation ideal $\ker \Pi_{\text{phys}}$ and a faithful representation of the quotient $\mathcal{A}_{\text{prim}} / \ker \Pi_{\text{phys}}$.

Proposition 8 (Compression of finite joint selection). *Once the primitive typed presentation is fixed, selection of the joint ideal, quotient representation and distinguished role images can be compressed to one coherent Π_{phys} , rather than a table of independently fitted cross-terminal gains.*

Proof. The kernel of a unital $*$ -homomorphism is a two-sided $*$ -ideal. The first isomorphism theorem identifies the quotient with its image. The images of the typed generators then determine the represented role packet and all algebraically generated mixed relations. $\square$

The proposition does not construct or occupy Π_{phys} . Distinct homomorphisms can preserve all currently fixed marginal and overlap data while changing mixed moments. A complete seed-owned relation presentation and a unique typed-unitary zero orbit would be sufficient, but are not derived here.

7.1 What Π_{phys} does and does not select

The homomorphism fixes algebraic relations and a represented quotient. It does not by itself fix a faithful state on that quotient. Conversely, the incidence/stiffness pair in Eq. (17) fixes a density operator only after the carrier and represented role algebra are known. The two data are complementary:

$$\Pi_{\text{phys}} : \text{relations and representation}, \quad (R_J, K_s) : \text{occupation and joint state}. \quad (24)$$

Calling them “equivalent” would require an additional common-source theorem that reconstructs each from the other. No such theorem is assumed here.

Proposition 9 (Faithful quotient representation). *If $\Pi_{\text{phys}} : \mathcal{A}_{\text{prim}} \rightarrow \mathcal{B}(\mathcal{H}_{\text{post}})$ is a unital $*$ -homomorphism, then the induced map*

$$\tilde{\Pi}_{\text{phys}} : \mathcal{A}_{\text{prim}} / \ker \Pi_{\text{phys}} \longrightarrow \text{im } \Pi_{\text{phys}} \quad (25)$$

is a faithful unital $$ -isomorphism.*

Proof. The kernel is a two-sided closed $*$ -ideal. The first isomorphism theorem makes the induced map bijective and $*$ -preserving; injectivity is precisely faithfulness on the quotient. $\square$

The choice of $\ker \Pi_{\text{phys}}$ remains physical information. The same primitive generator list can admit inequivalent ideals and hence inequivalent joint physics. Marginal agreement cannot select among them when the distinguishing relation lies in the marginal descriptor kernel.

8 Countermodels separating the logical levels

The preceding results are easiest to audit through finite controls.

Same marginals, different state. Equation (7) keeps both reduced density matrices fixed while changing one joint moment. Identifiability fails even though both joint states exist.

Same presentation, different quotient. Let $\mathcal{A}_{\text{prim}} = \mathbb{C}[x]/(x^2 - x)$, viewed as a unital $*$ -algebra with $x = x^*$. The characters $\Pi_0(x) = 0$ and $\Pi_1(x) = 1$ are both unital $*$ -homomorphisms but have different kernels. A generator name and its typing therefore do not select one physical relation ideal.

Same carrier dimension, different incidence rank. Take $K_s = I_2$, $R_{\text{onto}} = I_2$, and

$$R_{\text{deficient}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (26)$$

The first produces $G_J = I_2$; the second produces a rank-one Gram operator. Carrier dimension alone therefore does not imply faithful occupation.

Near-singular but onto incidence. For $R_\delta = \text{diag}(1, \delta)$, $0 < \delta \ll 1$, the joint state is faithful but $\kappa(G_J) = \delta^{-2}$. Exact identifiability can coexist with severe practical instability, which is why the singular-value ledger is part of the operational audit.

9 Worked end-to-end finite packet

We now run one packet from typed source data to represented mixed moments. It is deliberately elementary: its purpose is to expose every domain, rank and normalization used by the abstract theorems, not to model a physical Tau carrier.

Let the joint carrier and source-leg spaces both be $\mathbb{C}^2$, and set

$$K_s = \begin{pmatrix} k_0 & 0 \\ 0 & k_1 \end{pmatrix}, \quad R_\delta = \begin{pmatrix} 1 & 0 \\ 0 & \delta \end{pmatrix}, \quad k_0, k_1 > 0, \quad \delta > 0. \quad (27)$$

The represented role algebra is $\mathcal{A}_s = M_2(\mathbb{C})$, with typed Hermitian generators I, X, Y, Z and Hilbert–Schmidt reference pairing $\langle a, b \rangle_{\text{HS}} = \text{Tr}(a^\dagger b)$. The occupied faithful state will be generated by the incidence. Its Gram operator is

$$G_\delta = R_\delta K_s^{-1} R_\delta^\dagger = \begin{pmatrix} k_0^{-1} & 0 \\ 0 & \delta^2 k_1^{-1} \end{pmatrix}, \quad (28)$$

and therefore

$$\rho_\delta = \frac{1}{k_0^{-1} + \delta^2 k_1^{-1}} \begin{pmatrix} k_0^{-1} & 0 \\ 0 & \delta^2 k_1^{-1} \end{pmatrix}. \quad (29)$$

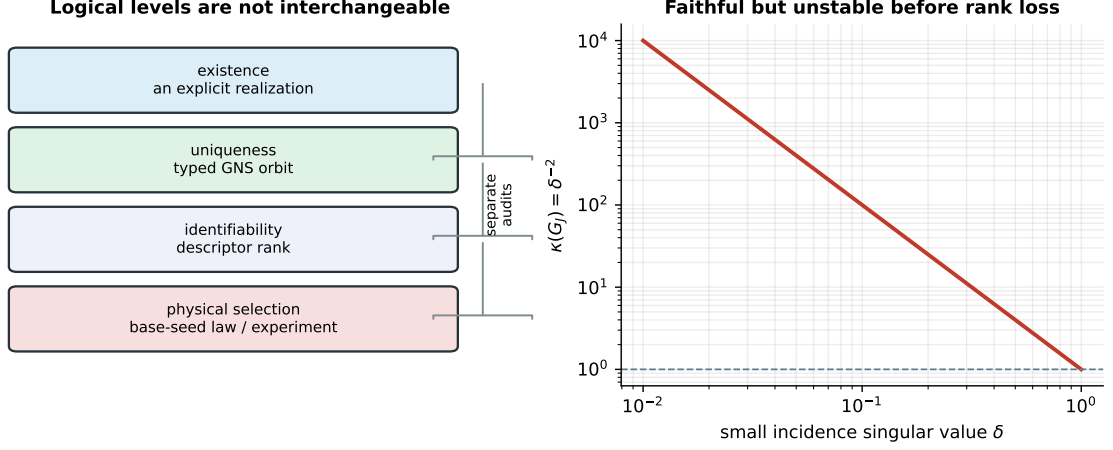


Figure 4: Left: existence, uniqueness, identifiability and physical selection require separate audits. Right: the worked packet stays faithful for every $\delta > 0$ while its Gram condition number diverges as rank loss is approached.

For every $\delta > 0$, R_δ is onto and ρ_δ is faithful. At $\delta = 0$, the second carrier direction lies in $\ker R_0^\dagger$, G_0 has rank one, and the full-carrier faithful-state claim fails exactly as Eq. (22) predicts. With $k_0 = k_1 = 1$,

$$\lambda_{\min}(G_\delta) = \delta^2, \quad \lambda_{\max}(G_\delta) = 1, \quad \kappa(G_\delta) = \delta^{-2} \quad (0 < \delta \leq 1). \quad (30)$$

The full Bloch descriptor

$$D_{XYZ}(\rho) = (\text{Tr } \rho X, \text{Tr } \rho Y, \text{Tr } \rho Z) \quad (31)$$

is injective on trace-one Hermitian matrices and reconstructs

$$\rho = \frac{1}{2}(I + D_X X + D_Y Y + D_Z Z). \quad (32)$$

For Eq. (29), $D_X = D_Y = 0$ and

$$D_Z = \frac{k_0^{-1} - \delta^2 k_1^{-1}}{k_0^{-1} + \delta^2 k_1^{-1}}. \quad (33)$$

Keeping only D_Z would identify this diagonal family after that family was frozen, but would not identify an arbitrary qubit state: the X and Y directions would remain in the descriptor kernel.

Finally take $\mathcal{A}_{\text{prim}} = M_2(\mathbb{C})$ with its Pauli presentation and choose

$$\Pi_{\text{phys}} = \text{id}_{M_2} : \mathcal{A}_{\text{prim}} \rightarrow \mathcal{B}(\mathbb{C}^2). \quad (34)$$

This is a unital $*$ -homomorphism with zero kernel, so it supplies the faithful quotient representation. The dephasing map $a \mapsto P_0 a P_0 + P_1 a P_1$ is unital and completely positive but not multiplicative on M_2 ; it cannot replace Π_{phys} in the quotient theorem even though it preserves the diagonal family.

The complete finite certificate for this toy is therefore

$$\mathfrak{C}_\delta = (M_2, \omega_\delta, K_s, R_\delta, D_{XYZ}, \Pi_{\text{phys}} = \text{id}, \mathcal{G}_{\text{eq}} = \{I\}, \text{occupation} = \rho_\delta). \quad (35)$$

where $\omega_\delta(a) = \text{Tr}(\rho_\delta a)$. It proves internal existence, typed GNS uniqueness, descriptor identifiability, onto-incidence faithfulness and representation consistency. It does *not* prove that a base-seed parent selects any member of this family, any value of k_0, k_1, δ , or the identity representation as a physical law.

9.1 Failure matrix for the worked packet

The same example separates failures that would otherwise look alike in an endpoint fit.

Modification	Mathematical result	Scientific diagnosis
$k_0 = k_1, \delta = 1$	$G = I, \rho = I/2, \kappa(G) = 1$	Symmetric, well-conditioned occupied packet; still no proof of physical selection.
$0 < \delta \ll 1$	$G > 0$ but $\kappa(G) = \delta^{-2} \gg 1$	Exact faithfulness survives, while finite precision makes the weak direction difficult to reconstruct.
$\delta = 0$	R is not onto and G has rank one	The claimed full-carrier faithful occupation is false; support reduction must be declared.
Retain D_Z only	Diagonal states are separated, arbitrary qubit states are not	The descriptor is complete only relative to a frozen commuting family; off-diagonal completion is unidentifiable.
Replace Π_{phys} by dephasing	The map is unital CPTP but not a *-homomorphism	It may define an irreversible record process, not the relation-ideal quotient used in Proposition 8.
Fit k_0, k_1, δ from the same terminal outcome	Many packets can reproduce one diagonal state	Endpoint agreement cannot establish upstream source ownership or select the physical parameter triple.

The rows give mutually different remedies. Rank loss requires a smaller support theorem; descriptor failure requires additional predeclared observables; nonmultiplicative dynamics require a CP/recovery analysis; and nature-selection failure requires an upstream source law or independent overdetermination. None is repaired by merely improving the fit of Eq. (29) to one measured density matrix.

10 Operational source-completeness audit

A finite source-signature claim should declare:

- (S1) the source algebra and true-null quotient;
- (S2) the faithful state or the incidence/stiffness pair generating it;
- (S3) all typed primitive roles and ownership multiplicities;
- (S4) the cross-sector observables required by the claim;
- (S5) the equivalence group and every non-quotiented automorphism;
- (S6) the rank test showing that retained observables separate the joint self-adjoint algebra; and
- (S7) whether existence, uniqueness and physical occupation are each proved, assumed or open.

Passing this audit proves informational completeness relative to the declared finite claim. It does not prove that Nature realizes the audited source.

10.1 A reproducible finite audit matrix

For a concrete model the declarations above reduce to finite linear algebra. The following matrix separates calculations from physical assumptions.

Object	Reproducible check	Failure meaning
Algebra packet	Verify associativity, unit and involution arrays in Eq. (2).	The proposed typed source is not a C^* -algebra presentation.
State packet	Diagonalize ρ_s on every declared occupied block.	A zero eigenvalue requires support reduction or a non-faithful theorem.
Descriptor	Compute rank D on the declared self-adjoint joint space.	Kernel directions are not identified by the retained data.
Incidence	Compute rank and singular values of R_J after true-null quotient.	Rank loss forbids faithful Gram occupation; small singular values signal instability.
Representation	Verify multiplicativity, unit and dagger preservation of Π_{phys} .	A merely unital linear or CP map does not define the claimed relation ideal by the first isomorphism theorem.
Equivalence	Test whether candidate solutions lie in the declared typed unitary orbit.	An undeclared automorphism is a second physical completion, not gauge by default.
Ownership	Trace every retained parameter to one source ledger entry.	A terminally inserted cross-gain violates source freeze.

The audit is deliberately compatible with negative results. A rank-deficient descriptor is not repaired by fitting its kernel directions to terminal data; it is reported as non-identifying. A non-onto incidence is not promoted by renaming its support the full carrier. A second zero orbit is not quotient gauge unless the equivalence group was declared before inspecting it.

11 Relation to established finite-dimensional results

The mathematical ingredients are standard; the contribution is their typed source-selection arrangement and explicit claim boundary. GNS gives a canonical cyclic representation after an algebra and state are fixed [1]. Stinespring characterizes completely positive maps through dilations, but does not select a physical source state or a unique nonminimal environment [2]. Quantum marginal theory asks whether reduced states are compatible with a common global state and when they determine it [3, 4, 5]. The present all-proper-marginal count is a direct tensor-basis kernel calculation within that broader problem.

What is specific to this program is the division of labor:

$$\text{typed source signature} \longrightarrow \left\{ \begin{array}{l} \text{minimal cyclic representation,} \\ \text{descriptor-rank audit,} \\ \text{incidence-generated occupation,} \\ \text{explicit nature-selection blocker.} \end{array} \right. \quad (36)$$

No novelty is claimed for GNS, Stinespring theory, or the marginal problem themselves. Nor does the finite analysis establish quantum gravity or the Standard Model. Its use is to prevent several terminal reconstructions from being mistaken for one physically selected common source.

12 What a physical source theorem must add

The remaining theorem cannot be another representation theorem conditional on the desired packet. It must provide a parent law whose occupied solutions jointly determine:

- (P1) the primitive typed algebra and its relation ideal;
- (P2) the carrier and the physical homomorphism Π_{phys} ;
- (P3) the source-leg stiffness K_s and incidence R_J ;
- (P4) a faithful occupied state or a declared support reduction;
- (P5) the equivalence group, including which automorphisms are gauge; and
- (P6) the terminally relevant descriptors without endpoint-dependent tuning.

An action principle could supply such data only if its variables and symmetry group are specified before variation and if its stationary orbit is unique up to the declared equivalence. A reconstruction from public terminal data can support or reject the rank, positivity and shared-parameter consequences of a candidate packet, but cannot by itself prove that the base-seed parent is its ontological source.

This distinction gives a finite reopening rule. The physical-selection branch is promoted only by either (i) a source theorem deriving (P1)–(P6), or (ii) a source-frozen experiment whose independent observables overdetermine the same packet and reject its allowed alternatives. Additional terminal fits without such overdetermination do not close the arrow.

12.1 Later source-completion status

The primary source-irredundant enriched completion now conditionally supplies one joint algebra/state/incidence packet of the required type, together with the typed quotient representation and the once-counted structural ledger. This closes internal nonemptiness for the adopted regular branch, not physical selection. An inequivalent completion with an independent base causal-work/energy-sharing carrier has the same current narrow source reduct. No selector using only that reduct can therefore establish that Nature occupies the primary packet.

13 What remains open

The internal alternatives are now finite:

$$\begin{array}{ll}
 \text{complete typed signature} & \Rightarrow \text{unique minimal cyclic representation,} \\
 \text{proper marginals only} & \Rightarrow \text{joint source non-identifiable,} \\
 \text{onto joint incidence} & \Rightarrow \text{faithful joint Gram state,} \\
 \text{unital *-homomorphism} & \Rightarrow \text{faithful quotient representation.}
 \end{array} \tag{37}$$

The Tau-specific blocker is the physical source of the third line and of Π_{phys} . Neither GNS uniqueness nor the dimension audit selects that source. Terminal fits cannot fill the gap without violating source freeze.

In particular, this paper does not prove:

- that the universal morphology seed exists or contains all primitive ordering information;
- that the physical base creates the declared finite algebra, faithful source state, onto incidence, or Π_{phys} as a uniquely Nature-selected packet; one enriched packet constructs them conditionally;

- that the incidence-generated state and the representation map descend from one microscopic once-counted functional;
- that terminal gravity, time, quantum, matter, gauge, or cosmological readouts occupy the finite roles used here;
- that the Standard Model spectrum, constants, generations, or a quantized spacetime geometry follow from the packet; or
- that any current data set empirically validates the Tau ontology.

The proper conclusion from a failed rank or occupation test is demotion of the candidate packet. It is not permission to add an independently fitted joint coefficient.

14 Conclusion

Finite source signatures separate four questions often conflated in foundation models: existence, representation uniqueness, identifiability and physical selection. Complete source data determine a unique minimal cyclic realization, but terminal marginals need not determine those source data. For n qudits, all proper marginals miss a full-weight subspace of dimension $(d^2 - 1)^n$; the four-qubit value is 81.

One onto source incidence with positive stiffness is sufficient to generate a faithful joint state, with conditioning controlled by the extremal singular values of the incidence. A coherent unital $*$ -homomorphism Π_{phys} separately compresses the ideal and representation choice. These are constructive finite targets, not nature-selection theorems. The primary enriched branch conditionally realizes the finite target listed in (P1)–(P6), but physical base-seed ownership and joint occupation remain unselected. The current-reduct counterpair proves that this is not removable by another representation theorem on the same narrow data.

The series handoff is therefore finite: Paper I supplies the architectural types, this paper supplies tests for source sufficiency and representation uniqueness, and Paper III begins only after such a packet and its occupied support have been frozen. Transport success cannot be used retroactively to infer the source packet, and source identifiability does not imply transport.

A Finite verification protocol

This appendix gives a model-independent calculation order for a proposed finite packet. It is intentionally mechanical so that a negative result can be reproduced.

Step 1: construct the algebra tables. Choose a basis containing the unit and store the arrays c_{ab}^c and j_a^b . Verify

$$c_{ab}^d c_{dc}^e = c_{bc}^d c_{ad}^e, \quad c_{1a}^b = c_{a1}^b = \delta_a^b, \quad (38)$$

$$(e_a e_b)^* = e_b^* e_a^*, \quad (e_a^*)^* = e_a. \quad (39)$$

A numerical tolerance is not an algebraic proof; exact or certified arithmetic is preferable when the entries permit it.

Step 2: freeze support and state. Construct G_s , diagonalize it on each central block, and declare whether zero eigenvalues are true nulls or failed occupation. Normalize only after this decision. Replacing a small eigenvalue by zero changes the claimed carrier and must be recorded as a model decision.

Step 3: assemble the descriptor. Vectorize a basis of traceless self-adjoint operators and write every retained expectation value as one row of D . Compute its exact rank when possible, then report σ_D in frozen physical units. A basis for $\ker D$ provides explicit countermodel directions.

Step 4: audit incidence. After removing only declared source nulls, compute the rank and singular values of R_J , then construct G_J . Check Eq. (22) directly and compare its eigenvalues with Eq. (19). This catches domain mistakes in $R_J K_s^{-1} R_J^\dagger$.

Step 5: verify the representation map. On primitive generators test unit, multiplication and dagger preservation. Compute the generated ideal and verify that the induced quotient map in Eq. (25) is injective. A CPTP or unital CP map may be physically useful, but it is not substituted for a $*$ -homomorphism in this particular quotient theorem.

Step 6: separate conclusions. Report existence, uniqueness, identifiability, conditioning and physical selection on separate lines. A pass in one line must not be copied into another. The minimal machine-readable certificate is therefore

$$\mathfrak{C}_s = (\Sigma_s, \text{rank } D, \sigma_D, \text{rank } R_J, \sigma_{\min}(R_J), \ker \Pi_{\text{phys}}, \mathcal{G}_{\text{eq}}, \text{occupation}). \quad (40)$$

B Proof-dependency ledger

The dependency graph is short enough to state without hiding assumptions:

- (D1) A finite unital C^* -algebra and faithful state imply a minimal GNS realization unique up to the declared typed unitary.
- (D2) Descriptor injectivity, not GNS uniqueness, determines whether retained data identify the joint state.
- (D3) Proper marginals fail this test whenever a full-weight tensor direction remains in the descriptor kernel.
- (D4) Positive stiffness and onto incidence produce a faithful joint Gram state; the singular values quantify stability.
- (D5) A unital $*$ -homomorphism selects a relation ideal and faithful quotient representation, but not its physical occupation.
- (D6) A base-seed source theorem must jointly own the data in (D1)–(D5) to close the Tau-specific nature-selection arrow.

No implication runs backward without extra hypotheses. In particular, faithful occupation does not imply onto incidence in every presentation, identifiability does not imply physical selection, and a faithful quotient representation does not determine a unique state.

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